

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Recalls the dimensions for energy and angular speed and forms an equation for dimensional consistency	AO1.1a	M1	$[E] = \text{ML}^2\text{T}^{-2}$ $[\omega] = \text{T}^{-1}$ $[I] = \frac{\text{ML}^2\text{T}^{-2}}{(\text{T}^{-1})^2} = \text{ML}^2$
	Completes a rigorous argument using both dimensions for energy and angular velocity to verify that the dimensions of I are ML^2	AO2.1	R1	
(b)	Uses dimensions to form a correct expression for the dimensions of $[I^\alpha W^\beta h^\gamma]$	AO1.1a	M1	$[I^\alpha W^\beta h^\gamma] = (\text{ML}^2)^\alpha (\text{MLT}^{-2})^\beta (\text{L})^\gamma$ $= \text{M}^{\alpha+\beta} \text{L}^{2\alpha+\beta+\gamma} \text{T}^{-2\beta}$ $\alpha + \beta = 0$ $-2\beta = 1$ $2\alpha + \beta + \gamma = 0$ $\alpha = 0.5$ $\beta = -0.5$ $\gamma = -0.5$
	Forms three simultaneous equations in three unknowns from 'their' $[I^\alpha W^\beta h^\gamma]$	AO1.1a	M1	
	Obtains correct values for α, β, γ	AO1.1b	A1	
	CAO			
				Total 5 marks

Q2.

LT^{-1}

For dimensions of u

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B1

$$LT^{-1} = M^\alpha L^\beta T^\gamma \times L^3 \times ML^{-3} \times LT^{-2}$$

M1 for equation with five components

M1A1

$$1 = \beta + 1$$

$$-1 = \gamma - 2$$

$$0 = \alpha + 1$$

Forming and solving equations (PI)

m1

$$\beta = 0, \alpha = -1, \gamma = 1$$

The dimensions of C are $M^{-1}T$

A1F

Alternative:

$$LT^{-1}$$

For dimensions of u

(B1)

$$LT^{-1} = C \times L^3 \times ML^{-3} \times LT^{-2}$$

M1 for equation with five components

(M1A1)

$$LT^{-1} = C \times L MT^{-2}$$

(m1)

The dimensions of C are $M^{-1}T$

(A1F)

[5]

Q3.

(a) $MLT^{-2} = \frac{[G]MM}{L^2}$

M1A1

$$[G] = L^3 M^{-1} T^{-2}$$

A1F

3

(b) $t = km^\alpha R^\beta G^\gamma$
 $T = M^\alpha L^\beta M^{-\gamma} L^{3\gamma} T^{-2\gamma}$

L, M, T for G are needed to gain M1

M1A1F

$$-2\gamma = 1 \Rightarrow \gamma = -\frac{1}{2}$$

$$\alpha - \gamma = 0 \Rightarrow \alpha = -\frac{1}{2}$$

$$\beta + 3\gamma = 0 \Rightarrow \beta = \frac{3}{2}$$

Getting 3 equations

m1

Solution

m1

Finding α, β, γ

A1F

Q4.

(a) $T^1 = L^a \times M^b \times (LT^{-2})^c$

M1A1

There is no M on the left, so $b = 0$

E1

3

(b) $T^1 = L^{a+c} \times M^0 \times T^{-2}$

M1

$$\begin{cases} -2c = 1 \\ a + c = 0 \end{cases}$$

equating corresponding indices

m1

$$a = \frac{1}{2}, c = -\frac{1}{2}$$

solution

m1

$$\therefore \text{Period} = k \frac{1}{g} g^{-\frac{1}{2}}$$

constant needed

A1F

4

[7]

Q5.

(a) (i) $ML^{-1}T^{-1}$

B1

1

(ii) $MLT^{-2}L^{-2} = ML^{-1}T^{-2}$

B2 for simplified form

B2,1

2

(b) $L^3 \times T^{-1} = L^{-1} \times L^a \times (MLT^{-2}L^{-2})^b \times (ML^{-1}T^{-1})^c$

M1A1F

$$= L^{-1} \times L^a \times M^b \times L^b \times T^{-2b} \times L^{-2b} \times M^c \times L^{-c} \times T^{-c}$$

$$= L^{-1+a-b-c} \times M^{b+c} \times T^{-2b-c}$$

PI

m1

$$\begin{cases} -1+a-b-c=3 \\ b+c=0 \\ -2b-c=-1 \end{cases}$$

Getting 3 equations

m1

Solution (finding at least one of a, b, c)

M1

$$a = 4, b = 1, c = -1$$

A1F

6

[9]

Q6.

- (a) Dimension of a force is $M L T^{-2}$

B1

Dimension of $\frac{mM}{r^2}$ is $M^2 L^{-2}$

Dimension of G is $\frac{MLT^{-2}}{M^2 L^{-2}}$

M1

$$= M^{-1} L^3 T^{-2}$$

A1

3

- (b) Inserting dimensions:
 $LT^{-1} = (M^{-1} L^3 T^{-2})^\alpha M^\beta L^{-\gamma}$

M1

$$= M^{\beta-\alpha} L^{3\alpha-\gamma} T^{-2\alpha}$$

A1✓

Equating terms in T; $\alpha = \frac{1}{2}$

M1

Equating terms in M; $\beta = \frac{1}{2}$

Equating terms in L; $\gamma = \frac{1}{2}$

cao

A1

4

[7]

Q7.

Dimensions of a and g are LT^2

B1

Dimensions of v is LT^{-1}

B1

$$\lambda = \frac{LT^{-2}}{(LT^{-1})^2}$$

M1

$$= L^{-1}$$

A1

[4]

Q8.

Dimension of a force is MLT^{-2}

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From Hooke's law; $\lambda = \frac{IT}{x}$

B1

$$= \frac{L \times MLT^{-2}}{L}$$

M1

$$= MLT^{-2}$$

A1

[4]

Q9.

$$\text{Force} = G \frac{m_1 m_2}{d^2}$$

or

$$G = \frac{d^2 \times \text{force}}{m_1 m_2}$$

Dimensions of a force (ma) is

MLT^{-2}

B1

$$\therefore \text{Dimensions of } G \text{ are } \frac{\text{L}^2 \cdot \text{MLT}^{-2}}{\text{M}^2}$$

M1

$$= \text{L}^3 \text{T}^{-2} \text{M}^{-1}$$

A1A1

[4]

Examiner reports

Q1.

Although students demonstrated a good understanding of dimensional notation, this question proved to be more challenging than expected, with a mean mark of 62% for part (a) and 59% for part (b).

In part (a) a number of students did not appear to know that $\frac{1}{2}$ was dimensionless. In addition, whilst many students knew the correct dimensions for energy, the dimensions of ω were often incorrectly stated.

In part (b), the majority of students were able to start with the correct line $[I^a W^b h^c] = (ML^2)^a (MLT^{-2})^b (L)^c$ but often did not then collect terms together correctly or subsequently failed to form three correct equations. The most successful students here always clearly stated their three equations before solving.

Q2.

Most candidates found this question straightforward. However, some candidates were not familiar with the usual notation for finding the dimensions of the physical quantities. Some candidates wrote MLT^{-2} for the dimensions of g . A small number of candidates were unable to form the correct equation of dimensions with the appropriate indices.

Q3.

This question was answered well by the great majority of candidates. A small minority of the candidates seemed to be unfamiliar with the concept of dimensional analysis. Some candidates used L instead of L^2 for the dimensions of x^2 and a few candidates used LT^{-2} instead of MLT^{-2} for the dimensions of the force. Some candidates benefited from follow-through marks in part (b).

Q4.

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The response to this question was quite variable. Too often, candidates were unable to answer this question correctly. Evidently, these candidates were not prepared for questions based on dimensional analysis. Some candidates' response to this question

was flawed as these candidates started with the formula $T = 2\pi \sqrt{\frac{1}{g}}$ and showed that $T = T$.

Q5.

This question was answered well by the great majority of the candidates. A small minority of the candidates seemed to be unfamiliar with the concept of dimensional analysis. Some candidates benefited from follow-through marks in part (b).

Q6.

Part (a) was well done; only a few candidates did not use MLT^{-2} as the dimension of force.

In part (b), most candidates knew the appropriate method but the manipulation of indices required was found to be challenging.

Q7.

Most candidates wrote down the correct dimensions of a , g and v . Unfortunately, a significant proportion had difficulty in obtaining the dimension of λ from these correct dimensions.

Q8.

This question was also answered well. A few candidates did not appreciate that the extension was a length and a few did not know the dimension of a force.

Q9.

Most candidates knew that the dimension of a force was MLT^{-2} , and they were then usually successful in completing the question.



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